TP 1: Returns and Random Walks

Statistics and Data Analysis for Financial Engineering, Chapters 1 and 2

Duration: 2 hours. Software: R.

Instructions

Every question must be answered with an R program. No calculation is done by hand. Write one script that runs from start to finish, and write your comments and conclusions in it.

The data come from R packages:

- **datasets** (included in R): *EuStockMarkets*, daily closing values of the DAX, SMI, CAC and FTSE indices, 1991–1998.
- **Ecdat**: *SP500*, daily log returns of the S&P 500, 1981–1991; *CRSPday*, daily net returns of GE, IBM, Mobil and the CRSP value-weighted index, 1989–1998; *Garch*, daily exchange rates of the US dollar, 1980–1987.
- **quantmod** (Exercise 5 only, needs internet): prices and dividends downloaded from Yahoo Finance.

Before each exercise, inspect the structure of the data set and read its help page.

Exercise	Topic	Time
1	Looking at the data	15 min
2	The normal distribution is not normal	15 min
3	Variances are not constant; risk	15 min
4	Net, gross, log and multi-period returns	20 min
5	Adjustment for dividends	10 min
6	Geometric random walk	15 min
7	Leverage and Monte Carlo	20 min
8	Testing the random walk model	10 min

Exercise 1. Looking at the data (EuStockMarkets)

"Always look at the data" is the first principle of Chapter 1.

(a) Plot the four indices. Describe the general trend of each and the period where they move the most.

(b) Compute the daily log returns of the four indices. Draw a scatter-plot matrix and compute the correlation matrix. Which two markets are the most correlated, and which market is the least correlated with the others?

(c) For each index, find the worst daily log return and the date on which it occurred. Do several markets share the same worst day? What event could explain it?

(d) An investor bought the DAX and sold the FTSE short on the first day. Using log returns only, plot the log of the ratio of the two investments over time. Did the DAX outperform the FTSE over the period?

Exercise 2. The normal distribution is not normal (SP500)

(a) Find the worst day in the sample. Give its log return and its net return, and identify the date.

(b) Standardise this return with the sample mean and standard deviation. Under a normal model, what is the probability of a day at least this bad? How many years would one wait, on average, for such a day (252 trading days per year)?

(c) Count the days whose standardised return exceeds 3, 4 and 5 in absolute value. Compare each count with the number expected under normality in a table.

(d) Repeat question (b) with a t-distribution with 4 degrees of freedom, rescaled to have variance 1. Compare the waiting times.

(e) Draw a normal QQ plot of the returns and describe where the data leave the line.

Exercise 3. Variances are not constant; measuring risk (CRSPday)

(a) For each of the four series and each year, compute the annualised volatility (standard deviation times $\sqrt{252}$). Present it as a table and plot it against the year. Is volatility constant over time? Which asset is the most volatile?

(b) Plot the autocorrelation function of the IBM returns and of the squared IBM returns. What do you conclude about independence and about volatility clustering?

(c) An investor holds \$1,000,000 of IBM for one day. Compute the 99% value-at-risk and expected shortfall of the loss in two ways: from the historical returns, and from a normal distribution with the sample mean and standard deviation. Which method gives the larger risk, and why?

Exercise 4. Net, gross, log and multi-period returns (CRSPday and Garch)

(a) Convert the net returns of the four series to log returns. What is the largest absolute difference between the two for each series? Is the approximation $\log(1 + R) \approx R$ good for daily data?

(b) Build the wealth of \$1 invested in each asset on the first day, using gross returns. Plot the four wealth paths on a log scale and give the final wealth of each.

(c) For GE, compute the monthly returns in two ways: as the product of daily gross returns, and as the sum of daily log returns. Check numerically that the two give the same monthly log return.

(d) Under the random walk model, the monthly log return has mean $k\mu$ and variance $k\sigma^2$, where k is the number of trading days in the month. Compare the sample mean and variance of the monthly log returns with these values. Comment on any difference.

(e) Compute the yearly net return of GE for each year. Which year was best and which was worst?

(f) In the Garch data set, the column dm is the price of one German mark in dollars. Its inverse is the price of one dollar in marks. Compute the daily net returns and the daily log returns of both series. Show that log returns change sign exactly when the quotation

is inverted, but net returns do not. Why does this make log returns more convenient for exchange rates?

Exercise 5. Adjustment for dividends (quantmod, internet required)

(a) Download the daily prices of Coca-Cola (KO) from 2015 to 2024 and its dividend history over the same period.

(b) Compute the return over the whole period in three ways: from the closing prices only, from the closing prices plus dividends using $1 + R_t = (P_t + D_t)/P_{t-1}$, and from the adjusted closing prices.

(c) How much of the total return comes from dividends? Is the adjusted-price return close to your dividend-adjusted return? Explain any remaining difference.

Exercise 6. Geometric random walk (EuStockMarkets)

(a) Estimate the mean μ and standard deviation σ of the DAX daily log returns.

(b) Starting from the first DAX value, simulate 1000 paths of a lognormal geometric random walk with these parameters, of the same length as the data. Plot 50 simulated paths together with the real DAX.

(c) What fraction of the simulated paths end above the real final value of the DAX?

(d) At several dates, compute the fraction of simulated paths whose log price lies within $\mu t \pm \sigma\sqrt{t}$ of the starting log price. Compare with the theoretical value.

(e) Compare the autocorrelation of the squared returns of the real DAX and of one simulated path. Which feature of the real data does the model miss?

Exercise 7. Leverage and Monte Carlo (CRSPday)

A hedge fund buys \$1,000,000 of the CRSP index with \$50,000 of its own capital and \$950,000 of borrowed money. If the value of the position closes below \$950,000, the fund must sell and loses all its capital.

(a) Estimate μ and σ of the daily log returns of the CRSP index.

(b) By simulation (10,000 paths, 45 trading days), estimate the probability that the position closes below \$950,000 at least once. Do it twice: with normal log returns, and by resampling the historical log returns with replacement (bootstrap).

(c) The fund now sells at the first close above \$1,100,000, at the first close below \$950,000, or after 100 trading days. With both methods, estimate the probability of a profit of at least \$100,000, the probability of a loss, the expected profit and the expected holding time.

(d) Do the normal and bootstrap results agree? Which one would you trust more, and why? Does the bootstrap capture volatility clustering?

Exercise 8. Is the lognormal random walk a good model? (EuStockMarkets)

Section 2.2.3 states that the model assumes normal and independent log returns.

(a) For each index, compute the skewness, the excess kurtosis and the p-value of the Jarque–Bera test of normality, where

$$JB = \frac{n}{6}\left(S^2 + \frac{K^2}{4}\right) \sim \chi^2_2 \text{ under normality.}$$

(b) For each index, compute the p-value of the Ljung–Box test with 10 lags on the log returns and on the squared log returns.

(c) Present all results in one table. Which assumption is rejected for all four markets? Is the model still useful? Discuss with the sentence "all models are false, but some models are useful".